

Evaluation of a State-Aware Conversational AI Agent for Immersive Anatomy Education

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Figure 1: (a) The user engages in dialogue with the AI tutor, which appears as a 3D holographic companion. (b) The user performs a phase-specific challenge.

ABSTRACT

The dynamic 3D morphology and temporal complexity of embryonic heart development make it difficult to comprehend. We introduce Reinheart, a conversational AI tutor for anatomy education that is state-aware and integrated into a mixed reality learning environment. The AI tutor uses OpenAI services to provide voice-based tutoring in real-time with interactive 3D visualization. Using a mixed-method analysis, ten anatomy experts assessed pedagogical performance, usability, and trust. The results indicated high ease-of-use and positive social presence, while qualitative feedback revealed limited uncertainty in communication and instability in challenge tasks. Experts supported the tutor's potential for conceptual reinforcement and guided exploration. To improve contextual reasoning and instructional reliability in MR anatomy learning, future work should extend this framework with domain-specific, uncertainty-aware AI.

Index Terms: Conversational AI, Virtual Reality, Embodied Virtual Assistants, Anatomy Education

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1 INTRODUCTION

A solid understanding of anatomy is fundamental to medical education, especially in the context of congenital abnormalities and development processes. The development of the embryonic heart is particularly difficult because it undergoes quick, multifaceted morphological changes in a short amount of time. The lack of temporal progression and spatial continuity in static 2D images, lectures, or cadaveric specimens makes it challenging to communicate these dynamics [1]. As a result, students frequently find it difficult to integrate successive developmental events, which impedes their ability to fully comprehend cardiac morphogenesis [2]. Immersive technologies like virtual, augmented, and mixed reality (VR/AR/MR) have become effective tools for tackling these challenges. They allow for stereoscopic visualization and interactive manipulation of 3D anatomical models [3]. Systems such as CardioGenesis4D show that temporal visualization in VR can improve understanding of anatomy and increase learner satisfaction [1]. However, even with better spatial and temporal understanding, most immersive applications depend on predefined content and provide limited instructional support once exploration starts [4, 5].

To address this issue, recent developments in generative artificial intelligence (AI) have introduced adaptive, conversational features that can respond to learner input in real-time. Conversational agents powered by large language models (LLMs) let learners ask questions and get context-aware explanations immediately [6, 7, 8]. Building on this, researchers have started investigating how these capabilities can be embodied in virtual agents that can lower cognitive load while maintaining immersion, especially when combined with interactive 3D visualizations [9].

Nevertheless, most AI-based VR learning environments in use today still rely on static conversation between AI agents and users

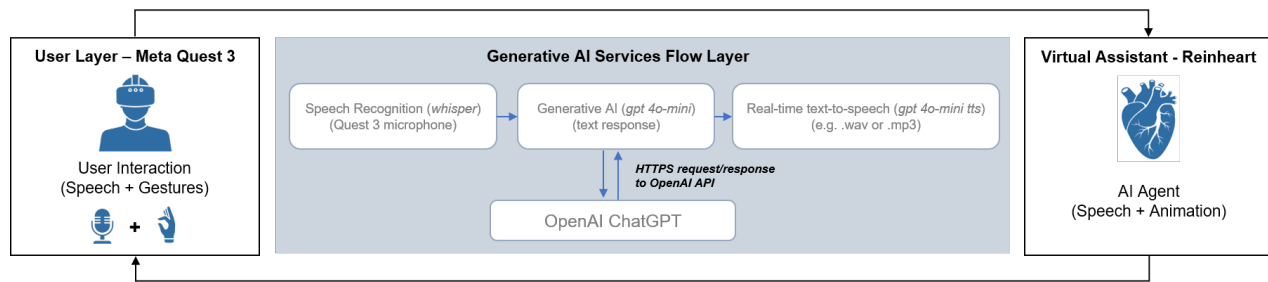


Figure 2: Dialogue flow of the AI tutor system, combining speech recognition, generative response, and animated delivery in an MR environment.

that are unable to adjust to the needs or context of specific learners. They often lack real-time awareness of user actions within complex spatial tasks, limiting their ability to provide meaningful, situated feedback. To fill this gap, we study how a conversational AI agent embedded in an interactive MR environment can facilitate context-aware, personalized learning in a spatially complex domain, the development of the embryonic heart. In particular, we investigate whether, in an anatomy education setting, such an embodied agent can improve learner experience, immersion, and perceived social presence (Figure 1 illustrates the typical interaction flow of the proposed system). The proposed AI agent improves current conversational AI approaches by adding real-time state awareness and spatial interaction tracking in a medical VR environment, which allows for specific guidance that matches user progress, closing the gap between fixed conversation systems and educational agents that respond to context. By placing the AI tutor directly into a procedural anatomy task, the system shows it can effectively provide structured training focused on a specific field instead of general conversation.

2 RELATED WORK

In anatomy education, mixed reality (MR) technologies have shown great promise, especially when it comes to explaining intricate spatial and temporal processes that are challenging to understand using conventional 2D materials [10]. Unlike static textbooks or flat screens, immersive environments improve depth perception, presence, and learner engagement. These factors are vital for understanding dynamic anatomical processes [10, 11]. Although MR offers better visualization, it has limited instructional functionality, especially for self-paced learning [7]. Students have been struggling without immediate instructor supervision, so current studies apply conversational AI to immersive systems. Such AI-based assistants, commonly using an LLM, have on-demand explanations, dynamic dialog responses, and personalized instructional support [12, 13].

In the area of anatomy education, Chheang et al. developed a ChatGPT-powered tutor that provided explanations sensitive to context during a VR exploration activity. They conducted a pilot experiment ($N = 16$), in which high usability and presence ratings on the avatar-based and screen-based systems significantly increased confidence with avatars [7]. Furthering this approach, various studies have employed the usage of LLMs within immersive learning for the medical or surgical field. Joseph et al. and Khakpaki demonstrated gains in accessibility and user engagement but noted a lack of real-time contextual awareness [14, 6], while Boubakri et al. and Shaban and Magzoub implemented domain-specific AI feedback modules [15, 16]. Despite these developments, most systems remain at the prototype stage, using general-purpose conversational frameworks without tracking user progress or spatial context.

The proposed system builds on this foundation by integrating a generative AI-based conversational tutor directly into an MR environment for embryonic heart development. Unlike earlier text-

based or post-hoc AI implementations, the proposed AI agent remains continuously aware of the tutorial phase and the user's location in 3D space, enabling context-sensitive, real-time guidance. The advantages of interactive embryonic heart models for enhancing learner outcomes have been shown in earlier research on spatial visualization systems [1]. These effects were confirmed across a variety of participant backgrounds, including those with little technical experience, in later large-scale studies [2].

The present study moves beyond visual comprehension to explore how embodied conversational agents can enhance trust, usability, and learning experience in MR anatomy instruction. The following sections describe the system architecture and the methodology used to evaluate its educational impact. The following sections outline the system architecture and evaluation methods of this integrated VR and AI approach.

3 MATERIALS AND METHODS

To investigate how a state-aware conversational agent can support immersive anatomy learning, we developed and evaluated an AI-integrated MR focused on embryonic heart development. This section outlines the system architecture, interaction model, participant sample, and the mixed methods study design used to assess usability, trust, and perceived social presence among medical experts.

3.1 System Architecture

We developed an interactive VR tutorial system using Unity (version 2022.3.26f1) and deployed it on a Meta Quest 3 headset. The application offers immersive 3D visualizations of how the embryonic heart develops. In this study, we added a state-aware AI tutor, "Reinheart", powered by OpenAI's multimodal *GPT-4o mini* framework, which combines *Whisper* for speech recognition, generative dialogue for creating responses, and *GPT-4o mini TTS* for voice output. Figure 2 shows the communication flow between the user, the Unity-based VR environment, and the OpenAI API services. The AI tutor stays aware of the user's state by tracking events in Unity. It logs user actions like object manipulation, gesture input, and phase transitions. These event signals help the AI tutor understand where the user is in the tutorial, he can then give tailored, task-specific feedback with real-time speech and animation synchronization. The interface, instructional content, and 3D assets stayed the same as in the previous iteration to keep consistency in the educational context and interaction design.

3.2 Interaction Model and Tutorial Flow

The tutorial consists of 12 sequential phases that represent key stages of embryonic heart development. Each phase is triggered by events and is aware of context, allowing AI tutor's guidance to adapt to user actions like grabbing, scaling, or pinching the 3D heart model. Users activate the tutor with a natural pinching gesture, keeping their hands free. Each phase includes (I) clear instructions, (II) an interactive task, (III) a short challenge, and (IV)

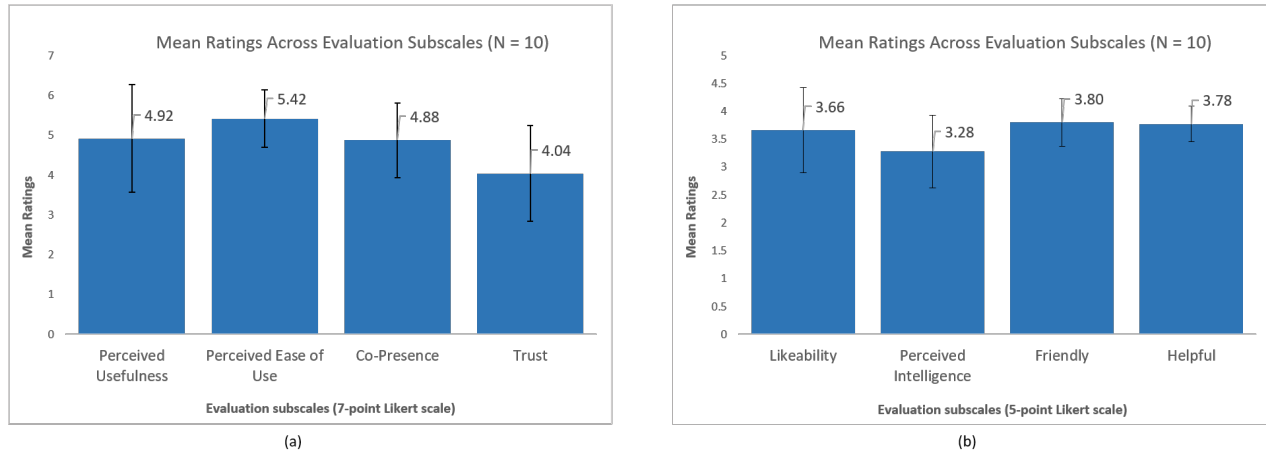


Figure 3: Mean ratings across (a) 7-point evaluation and (b) 5-point evaluation subscales ($N = 10$). Bar plots were used for clarity, given the small expert sample size. Error bars represent standard deviations.

instant feedback generated by AI. The design reflects a learning-by-doing approach that emphasizes continuity, spatial reasoning, and embodied thinking in 3D anatomy exploration. Inspired by Schott et al. [1, 2], the interaction model leverages natural hand gestures to directly manipulate the virtual heart model, reinforcing spatial intuition through embodied engagement.

3.3 Participants

Ten medical professionals ($N = 10$) specializing in anatomy from two different universities and medical faculties took part in the study ($M = 37.7$ years; 5 females, 5 males). Their backgrounds included postdoctoral researchers, anatomists, lecturers, and professors. Participants reported their experience with AI, VR, and MR technologies on a 5-point Likert scale. Six had prior exposure to VR-based anatomy applications. Informed consent was obtained from all participants prior to the study.

3.4 Study Procedure

Each session involved up to three participants in one room. After obtaining consent and providing a brief orientation, participants completed the AI tutor-guided MR tutorial at their own pace. Upon finishing, each participant individually completed a post-session questionnaire. Once all participants in the group had submitted their responses, they took part in a joint focus group interview. All sessions were audio-recorded for subsequent qualitative analysis. The study followed a within-subjects, exploratory design, with a focus on expert evaluation rather than comparative testing.

3.5 Evaluation Metrics

Participants completed standardized questionnaires that included: (I) Technology Acceptance Model (TAM) measures for perceived usefulness (PU) and perceived ease of use (PEOU) [17], (II) Trust in Automation (TiA) scales assessing reliability and dependability [18], (III) Co-Presence subscales from the Networked Minds framework [19, 20], and (IV) Perceived Social Intelligence (PSI) subscales assessing likeability, perceived intelligence, friendliness, and helpfulness [21]. PSI scales used 5-point Likert formats, while TAM, TiA, and Co-Presence used 7-point scales with acceptable internal consistency ($Cronbach's \alpha \geq 0.73$). Qualitatively, we thematically coded interview transcripts using reflexive analysis to identify recurring insights and suggestions for improvement [22].

4 RESULTS

4.1 Quantitative Analysis

Descriptive results based on the 7-point scale are shown to have a positive trend. The highest mean was found for PEOU at $M = 5.42$, followed by PU at $M = 4.92$, Co-Presence at $M = 4.88$, and Trust at $M = 4.04$. These results are shown graphically in Figure 3 (a). However, among the 5-point scale-based PSI measures, Friendliness and Helpfulness received the highest scores, at $M = 3.80$ and $M = 3.78$, respectively. On the contrary, the lowest score was found for Perceived Intelligence at $M = 3.28$ but likeability was a bit higher than it as $M = 3.66$, as shown in Figure 3 (b). It is important to note that the items for trust, helpfulness, friendliness, and likeability were explicitly linked to Reinheart's conversational style and voice persona, rather than the MR system interface. Descriptive statistics for the entire data set are presented in Table 1. Based on an expert sample, these results primarily provide exploratory insights.

4.1.1 Reliability Analysis

The reliability analysis indicated a strong internal consistency for most of the constructs. PU had a strong reliability estimate of 0.82, while PEOU had 0.79, with Trust also being strong with an estimate of 0.81. However, Co-Presence had an acceptable level of 0.76. Additionally on 5-point scales, Likeability had an estimate of 0.88, while Perceived Intelligence had an estimate of 0.81, which was strong. The two scales that were not measured for reliability were the Friendly and Helpful scales, which had only two items.

4.1.2 Correlation Analysis

Pearson's correlation analysis ($N = 10$) showed several prominent associations. The association between PU and PEOU was highly positive ($r = 0.71$), confirming the TAM, which suggests usability drives usefulness. The moderate association between PU and Trust was also evident ($r = 0.51$), suggesting functional perceptions impact reliability beliefs. Trust and perceived intelligence were highly interlinked ($r = 0.91$), confirming PSI theories and models about the association between superior intellect and building trust. The moderate association between likeability and Trust ($r = 0.43$) confirmed normal social-affective patterns, and a weak association between Co-presence and likeability ($r = 0.12$) indicated a potential divergence between social-affective experiences and shared presence in an AI tutor-based VR setting concerning experts, these relationships are visually summarized in Figure 5. The association between

Table 1: Descriptive statistics (Mean, SD, Min, Max) and reliability coefficients (Cronbach's α) for the 7-point and 5-point evaluation scales.

7-Point Scales					
Scale	Mean	SD	Min	Max	α
Perceived Usefulness	4.91	1.35	2.50	6.33	0.82
Perceived Ease of Use	5.41	0.71	4.50	6.50	0.79
Co-Presence	4.87	0.93	3.50	6.75	0.76
Trust	4.04	1.19	3.00	6.60	0.81
5-Point Scales					
Scale	Mean	SD	Min	Max	α
Likeability	3.66	0.76	2.60	4.80	0.88
Perceived Intelligence	3.28	0.65	2.40	4.60	0.81
Friendly	3.80	0.42	3.00	4.50	N/A
Helpful	3.78	0.32	3.25	4.25	N/A

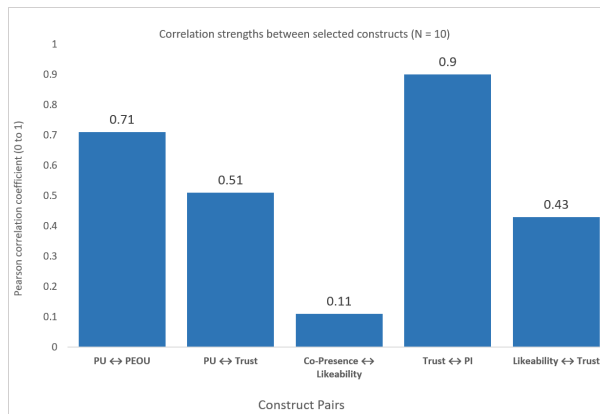


Figure 4: Pearson's correlation strengths between key evaluation constructs.

these variables reiterates theoretical predictions concerning the importance of usability and perceived intelligence in building Trust in AI tutors for students [23]. These correlations align with the TAM and PSI frameworks.

4.2 Qualitative Analysis

The feedback from experts was analyzed using reflexive thematic analysis [22, 24], revealing six key themes, i.e., Capabilities, Timing, Communicating Uncertainty, Explainability, Correction, and Learning Adaptivity. The experts found the AI tutor to be friendly and approachable. However, he lacked depth in concepts and understanding of context. According to one expert, "Reinheart feels approachable and confident, but sometimes answers as if it knows everything, even when it doesn't." The overall timing and flow of conversation were found to be acceptable, though at times inconsistent, showing signs of interruption and brevity. There were also concerns about the challenge phase, where irregularities in task registration affected the overall flow of learning, causing experts to repeat their actions. According to one expert, "When the challenge didn't register my gesture, it interrupted the flow, and the AI didn't respond to that error. Collectively, experts regarded the system as technically stable and engaging, with high usability and presence but lower trust and perceived intelligence. These findings suggest that while AI tutor effectively supports orientation and foundational learning, it requires substantial improvements in uncertainty communication, domain reasoning, and adaptive dialogue design before

being suitable for autonomous or advanced medical instruction.

5 DISCUSSION, LIMITATIONS, AND FUTURE WORK

Our findings demonstrate the feasibility and educational promise of integrating an AI-driven tutor into immersive anatomy learning. The quantitative analysis showed high levels of PEOU ($M=5.42$) and moderate levels of PU ($M=4.92$) in accordance with TAM findings. The high correlation coefficient of Trust and Perceived Intelligence ($r = 0.91$) with each other indicated that the users' belief in the intellect of the AI tutor played a large role in shaping their trust in the system, as is specified in models of social intelligence. The qualitative insights indicated that natural dialogue and embodiment were beneficial, but that a lack of transparency in corrections and inconsistent correction of errors decreased experts' confidence in the system. Lack of communication of uncertainty influenced overconfident answers, which decreased credibility [25]. This confirms the findings of social presence theory, stating that to achieve co-presence, interaction must be fluent and honest about knowledge [26, 27].

The primary limitation identified was the instability of challenge-based phases, where inconsistent task registration occasionally disrupted the learning flow and required users to repeat phases. Additionally, AI tutor's reasoning capability remained constrained by the general-purpose language model, limiting its ability to express uncertainty and provide nuanced explanations in specialized anatomy contexts. Dependence on cloud-based inference also introduced response latency, reducing conversational smoothness.

Addressing these issues, system stability, reasoning depth, and real-time performance should be critical for future scalability. Subsequent iterations should integrate a domain-specific, uncertainty-aware AI model fine-tuned on anatomical corpora, featuring adaptive dialogue aligned with learner progression, improved spatial and contextual understanding, and hybrid reasoning for greater reliability and coherence. Future evaluations should deploy these enhancements in authentic anatomy classrooms using long-term, mixed-methods assessment to examine usability and educational impact.

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